Background info

Calkoo introduced a simple online VAT calculator for calculating tax on their website:

<<http://www.calkoo.com/?lang=3&page=8>>

User story

As a traveller, I want to calculate the various purchase/tax amounts to verify that my shop receipt is correct, and the merchant correctly charged the VAT for my recently bought items.

ACCEPTANCE CRITERIA

- User must select a country which applies VAT scheme
- User must be able to choose a valid VAT rate for the selected country
- One of the following amounts are supported as an input:
  - Net
  - Gross
  - VAT
- Amounts can be entered with maximum 2 decimal digit precision
- Given all mandatory fields (country, vat rate, one of the amounts) are provided, the website will calculate and show the other 2 amounts which were not provided originally as an input value
- The API provides an error with meaningful error message, in case of:
  - negative input
  - amount >999.999.999

TECHNICAL REQUIREMENTS

- Please document test cases (manual) that come to your mind.
- In case you encounter bugs - which one and how would you report them (provide example)
- In case there are open questions regarding requirements/acceptance criteria:
  - Document the questions you would like to ask/clarify with End users/business/product owner based on the given information.
- Please identify at least two test cases for automation:
  - explain why you would choose specifically those
  - automate them using Cypress, Robot or Selenium (if having experience)

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• One of the following amounts are supported as an input:

o Net

o Gross

o VAT

• Amounts can be entered with maximum 2 decimal digit precision

• Given all mandatory fields (country, vat rate, one of the amounts) are provided, the website will calculate and show the other 2 amounts which were not provided originally as an input value

• The API provides an error with meaningful error message, in case of:

o negative input

o amount >999.999.999

TECHNICAL REQUIREMENTS

• Please document test cases (manual) that come to your mind.

Feature: Online VAT Calculator

Scenario 1: Calculate value-added tax and Price incl. VAT for given price without VAT.

Given: The user visits the VAT calculator website

When: The user selects a "<country>" that applies VAT scheme

And the user selects a "<validVatRate>" for the selected "<country>"

And the user enters the “<priceWithoutVat>”

Then: The website should calculate and display the <valueAddedTax> and <priceInclVat> Amounts for the given input

And value-added tax and Price incl. VAT should match the expected values for the given input.

Examples:

country	validVatRate	priceWithoutVat	valueAddedTax	priceInclVat
Austria	10%	83.33	16.67	100
Germany	19%	50	9.50	59.50
France	20%	120	24.00	144.00

Feature: Online VAT Calculator

Scenario 2: Calculate value-added tax and price without VAT for given Price incl. VAT.

Given: The user visits the VAT calculator website

When: The user selects a “<country>” that applies VAT scheme

And the user selects a “<validVatRate>” for the selected “<country>”

And the user enters the “<priceInclVat>”

Then: The website should calculate and display the “<valueAddedTax>” and “<priceWithoutVat>” Amounts for the given input

And value-added tax and Price without VAT should match the expected values for the given input.

Examples:

country	validVatRate	priceWithoutVat	valueAddedTax	priceInclVat
Austria	10%	83.33	16.67	100
Germany	19%	50	9.50	59.50
France	20%	120	24.00	144.00

Feature: Online VAT Calculator

Scenario 3: Calculate Price incl. VAT and price without VAT for given value-added tax.

Given: The user visits the VAT calculator website

When: The user selects a “<country>” that applies VAT scheme

And the user selects a “<validVatRate>” for the selected <country>

And the user enters the “<valueAddedTax> “

Then: The website should calculate and display the “<priceInclVat>” and “<priceWithoutVat>” Amounts for the given input

And value-added tax and Price without VAT should match the expected values for the given input.

Examples:

country	validVatRate	priceWithoutVat	valueAddedTax	priceInclVat
Austria	10%	83.33	16.67	100
Germany	19%	50	9.50	59.50
France	20%	120	24.00	144.00

Feature: Online VAT Calculator

Scenario 4: negative scenario. Invalid input for mandatory fields (negative and overflown values with boundary values).

Given: The user visits the VAT calculator website

When: The user selects a “<country>” that applies VAT scheme

And the user selects a “<validVatRate>” for the selected <country>

And the user enters a “<invalidValue>” for “<mandatoryField>”

Then: The website should display an “<errorMessage>”

Examples:

invalidValue	validVatRate	mandatoryField	errorMessage	country
1000000000	10%	value-added tax	"Invalid input"	Austria
1000000001	19%	price without VAT	"Invalid input"	Germany
-1	19%	value-added tax	"Invalid input"	Germany
-2	20%	price without VAT	"Invalid input"	France
-999999	20%	price incl. VAT	"Invalid input"	France
0	10%	value-added tax	"Invalid input"	Austria

Feature: Online VAT Calculator

Scenario 5: negative scenario. Invalid amount of decimal digit after point.

Given: The user visits the VAT calculator website

When: The user selects a “<country>” that applies VAT scheme

And the user selects a “<validVatRate>” for the selected <country>

And the user enters a “<moreDecimalDigitsValue>” for “<mandatoryField>”

Then: The text field should display only the first 2 decimal digits.

Examples:

moreDecimalDigitsValue	validVatRate	mandatoryField	errorMessage	country
12.345	10%	value-added tax	"Invalid input"	Austria
444.001	19%	price without VAT	"Invalid input"	Germany
6454.9999	19%	value-added tax	"Invalid input"	Germany
21232.43433	20%	price without VAT	"Invalid input"	France

Feature: Online VAT Calculator

Scenario 6: Verify that the VAT calculator component works correctly on different browsers and platforms and scales to different screen sizes.

Given: The user visits the VAT calculator website using <browser > and <platform> and <screenSize>

When: The user selects a “<country>” that applies VAT scheme

And the user selects a “<validVatRate>” for the selected “<country>”

And the user enters a “<validValue>” for “<mandatoryField>”

Then: The VAT calculator component should work correctly on each “<browser >” and “<platform>” and the user should be able to interact with the autogenerated graph as expected.

Examples:

validValue	validVatRate	mandatoryField	country	screenSize	browser	platform
12.34	10%	value-added tax	Austria	1366x768	Google Chrome	Windows
444.22	19%	price without VAT	Germany	1920x1080	Mozilla Firefox	Windows
6454.99	20%	price incl. VAT	France	768x1024	Apple Safari	MacOS
6454.99	20%	price incl. VAT	France	1024x768	Microsoft Edge	Windows
6454.99	20%	price incl. VAT	France	375x667	Opera	MacOS

Feature: Online VAT Calculator

Scenario 7: Verify the correct functionality of the reset button.

Given: User visits the VAT calculator website

When: User selects a “<country>” that applies VAT scheme

And User selects a “<validVatRate>” for the selected <country>

And User enters a “<validValue>” for “<mandatoryField>”

And User clicks on the reset button.

Then: The VAT calculator component should be reloaded and text fields should be cleansed and country and VAT rate should be set to the value by default.

Examples:

country	validVatRate	priceWithoutVat	valueAddedTax	priceInclVat
Austria	10%	83.33	16.67	100
Germany	19%	50	9.50	59.50
France	20%	120	24.00	144.00

Not fully manual scenarios

Scenario: Verify the VAT calculator can handle a high volume of requests.

Scenario: Verify the VAT calculator can handle many concurrent users.

Scenario: Verify the VAT calculator can handle peak loads.

Verify that the VAT calculator component is accessible to all users, including those with disabilities

- **In case you encounter bugs - which one and how would you report them (provide example)**

1. According to the AC: “Amounts can be entered with maximum 2 decimal digit precision.”

   Input fields should only allow users to enter numeric values with a maximum of two decimal digits after the decimal point, but in reality user can enter more than two digits after decimal point.

   Additionally, this criterion implies that any decimal value entered with more than two decimal digits should be rounded to two decimal digits before performing any calculations on the entered value. But this is not happening as can be seen on the following examples:

   https://i.imgur.com/utN529g.png

   https://i.imgur.com/WBMA2VR.png

   If that was the case, there should not be difference on the final calculated value Price incl. VAT
2. According to the AC: “The API provides an error with meaningful error message, in case of amount >999.999.999.”

   There is no alert for the user but instead the last digit of the overloaded number is just ignored.

   https://i.imgur.com/sTdQAcp.png
3. According to the AC: “The API provides an error with meaningful error message, in case of negative input”. When the user enters a negative value, there is no clear indication that this is

not a valid value (for example input field is not highlighted to indicate the error) and the message that appears is related to the graphic instead.

https://i.imgur.com/1AwwSvD.png

similarly for text inputs
https://i.imgur.com/GsyivMm.png

4. Automatically calculated fields are still editable by user, even if they are not selected to be considered with the checkbox and they appear greyed out.
https://i.imgur.com/l04izNA.gif

5. The site should not be reloaded when the user clicks on the "reset" button. Instead, only the input fields should be cleansed. Reloading the entire site would be unnecessary and could lead to a poor user experience.

   https://i.imgur.com/k2d7fxH.gif

6. Too low contrast in this component for accessibility purposes.
https://i.imgur.com/zY2lwhD.png

7. Weird animation when rolling over the percentage overlay (similar transparent layer on mobile).

   https://i.imgur.com/VZsNLkx.gif

8. Security Testing: The calculator does not use HTTPS, which means that the data sent between the user's browser and the server is not encrypted, leaving it vulnerable to interception and manipulation.

9. Recovery Testing: If the user accidentally closes the browser or loses the internet connection while using the calculator, any input data entered will be lost, and the user will have to start over.

In case there are open questions regarding requirements/acceptance criteria:
o Document the questions you would like to ask/clarify with End users/business/product owner based on the given information.

As the user history is a first approach to a desired functionality and it is expected for the specific requirements to be documented at later stages, this is enough as a start point. I would only add a clarification on the allowance of two digits after the point, because it is not clear whether the user can enter the whole number and then the figure is re calculated or if the number should be simply truncated while the user is typing in the number.

For later stages I would suggest further discussions on the specific details of the implementation, proposing questions as these among others:

1. There is no mention on the acceptance criteria about the warning the user should get in case of entering alphabetic characters or special characters, which obviously shouldn't be allowed.

2. Are there any specific accessibility requirements that need to be considered for the VAT calculator component, such as support for screen readers or keyboard-only navigation.

Please identify at least two test cases for automation:
o explain why you would choose specifically those

I would select test scenarios number 1,2 and 3. (Calculate value-added tax and price incl. VAT for given price without VAT, Calculate value-added tax and price without VAT for given Price incl. VAT. AND Calculate Price incl. VAT and price without VAT for given value-added tax)

These TC´s were selected due to the importance for the business on the correct functionality of the basic calculation behavior. Also, as this test cases cover the core functionality, they are expected to be run frequently, saving time and effort by automating repetitive and time-consuming tasks

o automate them using Cypress, Robot or Selenium (if having experience)

1. **Feature file:**

```
Feature: Online VAT Calculator

  Scenario: Calculate value-added tax and price incl. VAT for given price without VAT
    Given the user visits the VAT calculator website
    When the user selects a "<country>" that applies VAT scheme
    And the user selects a "<validVatRate>" for the selected "<country>"
    And the user enters the "<priceWithoutVat>"
    Then the website should calculate and display the "<valueAddedTax>" and "<priceInclVat>"
amounts for the given input
    And the value-added tax and price incl. VAT should match the expected values for the given
input

    Examples:
      | country   | validVatRate | priceWithoutVat | valueAddedTax | priceInclVat |
      | Austria   | 10%          | 83.33           | 16.67         | 100          |
      | Germany   | 19%          | 50              | 9.50          | 59.50        |
      | France    | 20%          | 120             | 24            | 144          |
```

Steps definitions file:

```
import { Given, When, Then } from 'cypress-cucumber-preprocessor/steps';
```

```
import VatCalculatorPage from '../pages/vat_calculator_page';

const vatCalculatorPage = new VatCalculatorPage();

Given('User visits the VAT calculator website', () => {
  vatCalculatorPage.visit();
});

When('User selects a {string} country', (country) => {
  vatCalculatorPage.selectCountry(country);
});

When('User selects a {string} VAT rate', (vatRate) => {
  vatCalculatorPage.selectVatRate(vatRate);
});

When('User enters the {string} price without VAT', (price) => {
  vatCalculatorPage.enterPriceWithoutVat(price);
});

Then('The website should calculate and display the {float} value-added tax amount and {float} price incl.
VAT amount for the given input', (expectedVatAmount, expectedTotalPrice) => {
  vatCalculatorPage.getVatAmount().should('eq', expectedVatAmount);
  vatCalculatorPage.getTotalPrice().should('eq', expectedTotalPrice);
});
```

Vat calculator page

```
class VatCalculatorPage {
  visit() {
```

```
  cy.visit('https://www.calkoo.com/en/vat-calculator');
 }

 selectCountry(country) {
  cy.get("select").select(1).invoke("val").should("eq", "1")
 }

 selectVatRate(vatRate) {
  const vatRateCheckbox = cy.get(`label[for="${vatRate}".css-label]`);
  vatRateCheckbox.click();
 }

 enterPriceWithoutVat(price) {
  const vatInformation = cy.get('label[for="F1"].css-label');
  vatRateCheckbox.click();
  cy.get('#NetPrice').type(price);
 }
enterPriceIncluidingVat(price) {
  const inclVat = cy.get('label[for="F3"].css-label');
  inclVat.click();
  cy.get('#Price').type(price);
 }
enterValueAddedTax(price) {
  const inclVat = cy.get('label[for="F2"].css-label');
  inclVat.click();
  cy.get('#VATsum').type(price);
 }
 getVatAmount() {
  cy.get('#VATsum').invoke('text').then(parseFloat);
```

```
  }
  getTotalPrice() {
    cy.get('#Price').invoke('text').then(parseFloat);
  }
getAmountNoVat() {
    cy.get('#NetPrice').invoke('text').then(parseFloat);
  }
}


export default VatCalculatorPage;
```

2. **Feature file:**

Feature: Online VAT Calculator

Scenario: Calculate value-added tax and price without VAT for given Price incl. VAT.

Given the user visits the VAT calculator website

When the user selects a "<country>" that applies VAT scheme

And the user selects a "<validVatRate>" for the selected "<country>"

And the user enters the "<priceInclVat>"

Then the website should calculate and display the "<valueAddedTax>" and "<priceWithoutVat>" Amounts for the given input

And value-added tax and Price without VAT should match the expected values for the given input.

Examples:

| country | validVatRate | priceWithoutVat | valueAddedTax | priceInclVat |

| Austria | 10% | 76.19 | 8.81 | 85 |

| Germany | 19% | 42.02 | 7.98 | 50 |

| France | 20% | 120 | 20 | 144 |

Steps definitions file:

```
import { Given, When, Then } from 'cypress-cucumber-preprocessor/steps';
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const vatCalculatorPage = new VatCalculatorPage();

Given('User visits the VAT calculator website', () => {
  vatCalculatorPage.visit();
});

When('User selects a {string} country', (country) => {
  vatCalculatorPage.selectCountry(country);
});

When('User selects a {string} VAT rate', (vatRate) => {
  vatCalculatorPage.selectVatRate(vatRate);
});

When('User enters the {string} price incluiding VAT', (price) => {
  vatCalculatorPage.enterPriceIncluidingVat(price);
});

Then('The website should calculate and display the {float} value-added tax amount and {float}
priceWithoutVat amount for the given input', (expectedVatAmount, expectedAmountNoVat) => {
  vatCalculatorPage.getVatAmount().should('eq', expectedVatAmount);
  vatCalculatorPage.getAmountNoVat().should('eq', expectedAmountNoVat);
});
```

3. **Feature file:**

Scenario: Calculate Price incl. VAT and price without VAT for given value-added tax

Given the user visits the VAT calculator website

When the user selects a "<country>" that applies VAT scheme

And the user selects a "<validVatRate>" for the selected "<country>"

And the user enters the "<valueAddedTax>"

Then the website should calculate and display the "<priceInclVat>" and "<priceWithoutVat>" amounts for the given input

And the value-added tax and price without VAT should match the expected values for the given input

Examples:

| country | validVatRate | priceWithoutVat | valueAddedTax | priceInclVat |

| Austria | 10% | 83.33 | 16.67 | 100 |

| Germany | 19% | 50 | 9.50 | 59.50 |

| France | 20% | 120 | 24.00 | 144.00 |

Steps definitions file:

```
import { Given, When, Then } from 'cypress-cucumber-preprocessor/steps';

import VatCalculatorPage from '../pages/vat_calculator_page';


const vatCalculatorPage = new VatCalculatorPage();


Given('User visits the VAT calculator website', () => {

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});


When('User selects a {string} country', (country) => {

  vatCalculatorPage.selectCountry(country);

});


When('User selects a {string} VAT rate', (vatRate) => {
```

```
  vatCalculatorPage.selectVatRate(vatRate);
});

When('User enters the {string} Value Added tax ', (price) => {
  vatCalculatorPage.enterValueAddedVat(price);
});

Then('The website should calculate and display the {float} price incl. VAT amount and {float}
priceWithoutVat amount for the given input', (expectedVatAmount, expectedTotalPrice) => {
  vatCalculatorPage.getAmountNoVat().should('eq', expectedAmountNoVat);
  vatCalculatorPage.getTotalPrice().should('eq', expectedTotalPrice);
});
```